# WAEL MELKI

**ICT Engineer | Data, AI & Machine Learning | 5G Networks**

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## PROFILE

Telecommunications engineer from SUP'COM (graduated with *Très bien*) specializing in the intersection of data science, applied machine learning, AI engineering, business analytics and 5G networks. Built end-to-end systems spanning multi-source data integration, SQL analytics, explainable predictive models, BI dashboards, MLOps and RF simulation. Seeking an ICT engineering role where network expertise and data/AI skills turn operational data into reliable decisions.

**Key results (Huawei simulation):** **658 priority contacts** of **2,629 eligible** · peak load **68.26%** vs. 80% limit · ROC-AUC ~**0.71**

## EXPERIENCE

**Huawei Northern Africa** **Feb 2026 – Jul 2026**
*Final-Year Engineering Intern | AI-Driven 5G FWA Planning* Tunis, Tunisia

- Built a Python/PostgreSQL decision pipeline integrating CRM, usage, service-consumption, coverage and RAN KPI data from **5 sources** into a 50-column analytical base table for **12,412 subscribers** and **731 radio cells**.
- Applied profiling, deduplication, outlier and referential-integrity checks; resolved systematic data-quality anomalies.
- Developed explainable FWA suitability scoring, K-Prototypes segmentation and a calibrated XGBoost churn model; simulated PRB-based capacity to evaluate migration scenarios and prioritize contacts.

**Netdata Innovation** **Jul 2025 – Aug 2025**
*Engineering Intern | 3D Radio-Coverage Simulation* GitHub Remote, France

- Developed a local Python application for Wi-Fi/5G coverage simulation over imported 3D scenes, with a modular four-layer architecture and configurable antennas and materials.
- Implemented free-space, material-aware and ray-tracing propagation models with a PyQt6/PyVista/VTK interface.
- Profiled workloads and worked on multiprocessing, caching and background execution to keep simulations responsive.

## SELECTED PROJECTS

**Non-Invasive Diabetes Detection** SUP'COM tutored project, 2025–2026 | Python, scikit-learn GitHub

Joint research prototype for type 1 diabetes detection: SVM on ECG heart-rate-variability features and Random Forest on respiratory features, with subject-aware validation and late fusion (test accuracy **80.8%**, AUC **0.74**; 29 subjects, not clinical).

**Energy Analytics and Anomaly Detection** P2M project, May 2025 | Python, PyWavelets, Streamlit GitHub

Joint project that compresses appliance power signals with coif5 wavelets, detects anomalies with Isolation Forest and HDBSCAN and exposes results in an interactive Streamlit platform with forecasting.

**CIFAR-10 MLOps Platform** ML engineering | ZenML, MLflow, DVC, Docker Compose GitHub

Combines ZenML pipelines, MLflow tracking/registry, DVC versioning, MinIO, PostgreSQL and Docker Compose with Evidently drift detection and guarded automatic retraining.

**Chronos Agent** AI agent MVP v0.1 | FastAPI, LangGraph, Next.js GitHub

Orchestrates calendar actions through LangGraph tools with human confirmation, audit/undo logic, Google OAuth/Calendar and PostgreSQL; supports multiple LLM providers and Docker Compose deployment.

## SKILLS & INTERESTS

**Data & ML:** Python, SQL, C++, MATLAB, Pandas, NumPy, SciPy, scikit-learn, XGBoost, SVM, Random Forest, Keras
**AI & MLOps:** MLflow, ZenML, DVC, Evidently, LangGraph, FastAPI, Docker Compose, Git, Linux, pytest
**Databases & BI:** PostgreSQL, SQL Server, Power BI, DAX, star schema, ETL, data validation
**Wireless & Signals:** 5G FWA, RAN KPIs, PRB capacity, RF propagation, wavelets, ECG/respiratory signal processing
**Languages:** Arabic (native), French (fluent), English (fluent)
**Interests:** Competitive programming and algorithmic problem solving (92 problems solved on Codeforces)

## EDUCATION & TRAINING

**SUP'COM – Higher School of Communications of Tunis** **2023 – Sep 2026**
*National Engineering Diploma in Telecommunications* Tunis, Tunisia

Graduated with *Très bien*; final-year project: **16/20**. Focus: mobile networks, signal processing, cloud, embedded systems.

**IPEIT – Preparatory Institute for Engineering Studies of Tunis** **2021 – 2023**
*Preparatory Cycle in Mathematics and Physics | National engineering entrance exam:* ***63rd of 920*** *(top 7%)* Tunis, Tunisia

**Certifications & training:** TOEIC; MathWorks Machine Learning Onramp and MATLAB Onramp (online courses)